5COSC020W DATABASE SYSTEMS COURSEWORK

(2025/2026)

Lecturer Name : Nilumi Perera

Name : Dahamsee Kasthuriarachchi

UOW ID : w2120869

IIT ID : 20231314

Tutorial Group : A (BIS)

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Contents

Acknowledgment 2
List of Figures 4
List of Tables 4
Question 1 – Critical Evaluation 5
A. Weaknesses in the AI generated EERD 5
B. Strengths in AI generated EERD 6
Business Questions 7
Assumptions 8
Question 2 – Decision Log 9
Question 3 – Entity Data Dictionary Table 11
Question 4 – Specialization Data Dictionary Table 14
Question 5 – Relationships and Multiplicities Data Dictionary table 16
A. Binary Relationships 16
B. Ternary Relationships 18
Question 6 – Attributes and Primary key Data Dictionary Table 20
A. General and specialized entities and their attributes 20
B. Relationship Attributes 26
Question 7 – Conceptual EERD 28
Question 8 – Mapping Conceptual EERD to Logical ERD 29
Question 9 – Mapping Log 30

List of Figures

Figure 1 Business Questions 7
Figure 2 Conceptual EERD 28
Figure 3 Logical ERD 29

List of Tables

Table 1 Decision Log 10
Table 2 Entities 13
Table 3 Specializations 15
Table 4 Binary Relationships & Multiplicities 18
Table 5 Ternary Relationships & Multiplicities 19
Table 6 General and Specialized Entities & their Attributes 26
Table 7 Relationship Attributes 27
Table 8 Mapping Log 32

PART A

Question 1 – Critical Evaluation

A. Weaknesses in the AI generated EERD

1. **Missing basket entity.**
   The ERD connects Member directly to Order, skipping the Basket stage. According to the scenario, each member must first create a named basket before placing an order. Omitting the Basket entity prevents modelling saved or abandoned baskets and breaks the required workflow where an order originates from a basket.

2. **Incomplete product specialization.**
   The diagram includes only one specialization level Perishable vs Non Perishable, omitting the deeper hierarchy Product from Perishable -Food, Drink- Soft Drink, Alcoholic Drink. This simplification ignores explicit business rules distinguishing food from drink and alcoholic from non-alcoholic beverages. It prevents correct enforcement of constraints such as ABV for alcoholic drinks and allergen/nutritional info for food.

3. **Payment methods not modelled as specialization.**
   Payment is represented as a single entity instead of specialized subtypes for CardPayment and Mobile Payment. The brief specifies only two distinct payment modes, each with potentially unique attributes card number/expiry vs mobile wallet ID. Without specialization, those attributes cannot be captured cleanly, and validation/business rules become harder to enforce.

4. **No link between warehouse workers and orders they prepare.**
   Warehouse workers are not connected to the orders they prepare. The scenario requires traceability and accountability of staff during order fulfilment. Without this relationship, the system cannot identify which worker processed a particular order or track performance/responsibility.

B. Strengths in AI generated EERD

1. **Identification of core entities (Store, Product, Member, Order)**
   The model correctly recognizes the main real-world objects defined in the scenario, stores, products, members, and orders. These represent the essential data pillars of this business process. Their inclusion ensures the conceptual model captures the key operations of product offering, customer interaction, and order fulfilment.

2. **Inclusion of a relationship between Store and Product**
   The ERD establishes a link between Store and Product, acknowledging that each store maintains its own stock and pricing. This reflects the business model where multiple stores sell the same products at different prices and quantities. Modelling this association supports store-specific inventory and price management.

3. **Inclusion of product specialization (Perishable vs Non-perishable)**
   The diagram introduces specialization of Product based on perishability, aligning with the business rule that some products have expiry constraints. This demonstrates awareness of hierarchical data modelling and sets the foundation for further sub classification (Food vs Drink). Even though deeper levels could be added, the presence of the first specialization layer is conceptually sound.

4. **Use of primary keys and identifiers across entities**
   Each entity contains a clear unique identifier (storeID, productID, memberID). This facilitates entity integrity and enables the model to map cleanly to a logical schema later. Including these identifiers at the conceptual stage shows understanding of key constraints and supports data consistency.

Business Questions

Dahamsee Kasthuriarachchi
Oct 16, 2025 8:11 PM

UOW ID : W2120869

IIT ID : 20231314

GROUP : BIS GROUP C

1. How is stock level updated? Automatically post-order or manually by store?
2. Can stores have discounts or promotional pricing on certain items?
3. Can members rate or review products or stores?
4. Can members schedule delivery at a specific time?
5. How long can a shopping basket remain active before being cleared or expired?

Reply

Figure 1 Business Questions

Assumptions

1. **Product and Basket** - Some products may still not be added to any basket by anyone. They will remain in the store's stock.
2. **Collection and Order** - A collection is created when more than one order is ready to dispatch together.
3. **Collection and Vehicle** - Each collection is handled by exactly one vehicle. A vehicle may handle many collections over time. There is no collection exchange between the process of picking the collection from warehouse to delivering all the orders in it to all the customers at the end.
4. **Store and Product** - Each store offers multiple products and product price and stock level are store specific.
5. **Basket and Product** - A basket can include products from different stores.
6. **Basket and Order** - A basket can produce one order only. Each order originates from exactly one basket.
7. **Order and Payment** -Each order can be paid by one payment only and also the customer can use only one type of payment to pay for the entire order.
8. **Order and Collection** - An order can be included in one collection only.
9. **Vehicle and Driver** - Vehicles are temporary assigned to the drivers.
10. **Delivery and Vehicle** - Each delivery uses only one vehicle and each vehicle can carry out either one or multiple deliveries.

Question 2 – Decision Log

No.	Modelling Decision	Justification
1	Added a relationship attribute between 'w2120869_Product' and 'w2120869_Basket'.	A basket contains multiple products, and each product can appear in many baskets. The Foodz@Yourz system records specific information for each basket line — namely the *quantity* of the product selected and the *line_status* (e.g., ongoing or confirmed). Since these details describe the association between a particular basket and a particular product (not the entities themselves), *quantity* and *line_status* are modelled as relationship attributes.
2	Added a relationship attribute between 'w2120869_Store' and 'w2120869_Product'.	Between the entities 'w2120869_Store' and 'w2120869_Product', each product is stored in one or many stores. And as for stores, they can carry many products. So the attributes 'stock_Level' and 'price' are store dependent and they belong into the relationship between the two entities.
3	1..1 optional relationship between 'w2120869_Order' and 'w2120869_Basket'	A basket may optionally connects into an order because not all baskets becomes orders. Some are abondoned, some are cancelled. However every order originates from a basket.
4	Added 'w2120869_Delivey' and 'w2120869_Collection' as separate entities.	'w2120869_Delivery' and 'w2120869_Collection' are modelled independently due to distinct attributes. Combining them would cause matters for each fulfillment. Separate entities ensures clean modelling for the two entities.
5	Sorted the relationship and the multiplicities between the entities, 'w2120869_Order' , 'w2120869_Delivery' and 'w2120869_Collection'.	The system/ scenario states that and order cannot be both delivered and collected. And it cannot be splitted. Separating them into two different relationships ensures a clear separation of fulfillment types.
6	Added ternary relationship between the entities 'w2120869_Vehicle',	The relationships between the three entities is modelled in a ternary relationship because, a delivery can only

	‘w2120869_Driver’ and ‘w2120869_Delivey.’	occur when all three components exist and operate together. This will ensure that model accurately reflects operational constraints such as, vehicle allocation and delivery scheduling.
7	Added a ‘1..M’ relationship between ‘w2120869_Member’ and ‘w2120869_Basket’.	A single member may create multiple baskets over time because each basket represents a separate shopping activity. However they all belong to one single member. The baskets capture separate independent shopping sessions with different time stamps , products and purposes.
8	Modelled ‘w2120869_Product’ as generalization of the two specialized entities ‘w2120869_Perishable_Product’ and ‘w2120869_Non_Perishable_Product’.	It is stated in the scenario that certain things are perishable, and the responses to the business questions indicate that not all of the products are perishable, indicating that some are non-perishable as well.
9	Between the ‘w2120869_Worker’ generalization and specialization used ‘Optional’ as the participation constraint.	According to the case study, "drivers" and "ware house workers" are part of the internal workforce. The addition of "Optional" indicates that any employee who does not fit the definition of "driver" or "ware house worker" are still employed such as admin staff and cleaning staff.
10	Between the ‘w2120869_Perishable_Product’ generalization and specialization used ‘Optional’ as the participation constraint.	According to the clarifications the perishable products are not just food products or drinks. They could be things such as toiletries which still has an expiry date and is perished with time.

Table 1 Decision Log

Question 3 – Entity Data Dictionary Table

No.	Entity	Explanation
1	w2120869_Member	This entity represents all the registered Foodz@Yourz users, each with unique credentials and contact details who can create baskets, place orders and get them delivered to their home address.
2	w2120869_Basket	This entity represents the baskets created by members, each basket with unique identification which belongs to an individual member. This basket contains products from different stores and when members are satisfied with their selection, they can convert it into an order. If not it can be archived for further reference.
3	w2120869_Product	This entity represents the items offered by the stores which maybe perishable or non perishable. This also stores descriptive details about the products.
4	w2120869_Perishable_Product	This subtype specialized entity represents products that has expiry/nutritional info.
5	w2120869_Non_Perishable_Product	This subtype specialized entity represents the products that does not expire.
6	w2120869_Food	This subtype specialized entity represents edible food items consumed for nutrition.
7	w2120869_Drink	This subtype specialized entity represents This includes both alcoholic and non-alcoholic drinks and requires storage of attributes such as volume, brand.
8	w2120869_Alchohol	This subtype specialized entity represents the drink products that contain alcohol.

9	w2120869_Soft_Drink	This subtype specialized entity represents non alcoholic drinks such as sodas, juices or flavored water.
10	w2120869_Store	This entity represents the retail participants offering specific products with unique stock & price which participate or do not participate in Foodz@Yourz program.
11	w2120869_Order	This entity represents the confirmed requests derived from the baskets, each order derived from a single basket which can be archived for further reference and is handled by a single warehouse worker and belong to a single collection for sure.
12	w2120869_Payment	This entity represents the records of the payment details for orders with the payment status being the confirmation of an order.
13	w2120869_Card_Payment	This subtype specialized entity represents the payments for orders paid via credit or debit cards.
14	w2120869_Mobile_Payment	This subtype specialized entity represents the payments for orders made using mobile application.
15	w2120869_Collection	This entity represents the grouping of orders which has more than one order in each collection for pickup and warehouse handling.
16	w2120869_Delivery	This entity represents the transportation records of orders to members' addresses, each delivery being handled by a single driver and a single vehicle.
17	w2120869_Vehicle	This entity represents the transportation modes used by drivers for deliveries of collections or for other work related activities which is assigned to drivers temporarily.

18	w2120869_Two_Wheel_Vehicle	This subtype specialized entity represents vehicles with two wheels and have limited capacity.
19	w2120869_Four_Wheel_Vehicle	This subtype specialized entity represents vehicles with four wheels with high capacity.
20	w2120869_Worker	This entity represents Foodz@Yourz staff which can be warehouse workers, drivers or admin staff and other staff.
21	w2120869_Warehouse_Worker	This subtype specialized entity represents the workers who prepare orders.
22	w2120869_Driver	This subtype specialized entity represents the workers who deliver the orders and drive the vehicles for any kind of work related activities.

Table 2 Entities

Question 4 – Specialization Data Dictionary Table

No.	General Entity	Specialized Entities	Explanation
1	w2120869_Product	'w2120869_Perishable_Product' 'w2120869_Non_Perishable_ Product'	Perishable products has expiry, nutrition, allergens and non-perishable products has handling info. With the constraint being {Mandatory,Or} , the participation 'Mandatory' suggests that the products can only be categorized under perishable or non perishable products. The disjoint 'Or' suggests that there is no overlapping between these two categories, product cannot be categorized under both perishable or non perishable.
2	w2120869_Perishable _Product	'w2120869_Food' 'w2120869_Drink'	Perishable products can be foods or drinks. With the constraint being {Optional,And}, the participation 'Optional' suggests that there are perishable products other than foods and drinks such as toiletries. The disjoint 'And' suggests that there are products which can be both a food item and a drink such as yoghurt drink with an overlapping between these two sub classes.
3	w2120869_Drink	w2120869_Alcohol w2120869_Soft_Drink	Soft drinks have sugar level; alcoholic drinks have alcohol percentage and legal age. With the constraint being '{Mandatory,Or} , the participation 'Mandatory' suggests that the drinks can only be categorized under alcohol or soft drinks. The disjoint 'Or' suggests that there is no overlapping between these two categories as there cannot be a drink which can both be an alcoholic drink and a soft drink.
4	w2120869_Payment	w2120869_Card_Payment w2120869_Mobile_Payment	Superclass payment is specialized into card payment and mobile payment. With constraint being '{Mandatory,Or}' , the

			participation 'Mandatory' suggests that there are only two types of payment methods as card and mobile payments, eliminating payment types such as cash upon delivery. The disjoint 'Or' suggests that a particular order's payment can only be fulfilled by either one of these two categories with no overlapping between them.
5	w2120869_Vehicle	w2120869_Two_Wheel_Vehicle w2120869_Four_Wheel_Vehicle	Distinction based on capacity/number of wheels. With constraint being '{Mandatory,Or}' , the participation 'Mandatory' suggests that there is only two types of vehicles in this service platform which is either two wheel or four wheel and the disjoint 'Or' suggests that there is no overlapping between these two categories as a vehicle cannot be categorized under both of these vehicle types.
6	w2120869_Worker	w2120869_Warehouse_Worker w2120869_Driver	Workforce roles with distinct attributes and functions. With the constraint being '{Optional,Or}, the participation 'Optional' suggests that there can be workers with separate roles other than workhouse workers and drivers such as admin staff and cleaning staff. The disjoint 'Or' suggests that there is no worker that can be categorized under both of these subclasses as the roles requires very different skill sets.

Table 3 Specializations

Question 5 – Relationships and Multiplicities Data Dictionary table

A. Binary Relationships

No.	Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity	Explanation
1	w2120869_Member	1..1	creates	w2120869_Basket	0..M	Member can create minimum zero baskets.
						Member can create many baskets at most.
						Basket can belong to minimum one member.
						Basket can belong to only one member at most.
2	w2120869_Basket	0..M	includes	w2120869_Product	1..M	Basket includes at least one product.
						Basket includes many products at most.
						Product can belong to no basket at all.
						Product can belong to more than one basket.
3	w2120869_Product	1..M	offers	w2120869_Store	1..M	Store offers minimum one product.
						Store offers many products at most.
						Product can belong to minimum one store.
						Product can belong to many stores at most.
4	w2120869_Basket	1..1	converts to	w2120869_Order	1..1	Basket can be converted to minimum one order.
						Basket can be converted to just one order at most.
						Order can be created by minimum one basket.

						Order can be created by just one basket at most.
5	w2120869 _Order	1..1	paid by	w2120869 _Payment	1..1	Order is paid by minimum one payment.
						Order is paid by just one payment at most.
						Payment can belong to minimum one order.
						Payment can belong to just one order at most.
6	w2120869 _Order	M..M	include s	w2120869 _Collection	1..1	Order is included in minimum one collection.
						Order is included in just one collection at most.
						Collection can have more than one order.
						Collection can have many orders at most.
7	w2120869 _Collection	1..M	schedul ed for	w2120869 _Delivery	1..1	Collection is scheduled for minimum one delivery.
						Collection is scheduled for a single delivery at most.
						Delivery can have minimum one collection.
						Delivery can have many collections at most.
8	w2120869 _Vehicle	1..1	carries	w2120869 _Collection	1..M	Vehicle carries minimum one collection.
						Vehicle carries many collections at most.
						Collection can be carried by minimum one vehicle.
						Collection can be carried by just one vehicle at most.
9	w2120869 _Driver	1..M	uses	w2120869 _Vehicle	1..M	Driver uses minimum one vehicle.
						Driver uses many vehicles overtime.

						Vehicle can be allocated to minimum one driver.
						Vehicle can be allocated to many drivers overtime.
10	w2120869 _Order	1..M	process ed by	w2120869 _Warehous e_Worker	1..1	Order is processed by minimum one warehouse worker.
						Order is processed by just one warehouse worker at most.
						Warehouse worker can process minimum one order.
						Warehouse worker can process many orders at most.

Table 4 Binary Relationships & Multiplicities

B. Ternary Relationships

No.	Entity Name	Multip licity	Entity Name	Multip licity	Relat ions hip	Entity Name	Multip licity	Explanation
1	w2120869 _vehicle	1..1	w2120869 _Driver	1..1	uses	w2120869 _Delivery	1..M	A vehicle is used by just one driver to deliver minimum one to many deliveries at most at one instance.
2	w2120869 _Delivery	1..M	w2120869 _Vehicle	1..1	uses	w2120869 _Driver	1..1	A single delivery is delivered by using just one vehicle by one driver at most at a single instance.

3	w2120869 _Driver	1..1	w2120869 _Delivery	1..M	uses	w2120869 _Vehicle	1..1	A driver delivers one to many deliveries using one vehicle at most at a single instance.

Table 5 Ternary Relationships & Multiplicities

Question 6 – Attributes and Primary key Data Dictionary Table

A. General and specialized entities and their attributes

No.	Entity Name	Attributes with Primary Key	Explanation
1	w2120869_Member	w2120869_memberId {PK}	This attribute uniquely identifies each member which is why it is the primary key.
		w2120869_memberName	This attribute records the full name of each member.
		w2120869_memberAddress	This attribute records the address of each member for the delivery purposes.
		w2120869_memberTelNo	This attribute records the telephone number of each member in order to contact them.
2	w2120869_Basket	w2120869_basketId {PK}	This attribute uniquely identifies each basket which belongs to only one member which makes it the primary key.
		w2120869_basketName	This attribute records the name given to the basket by the member who created it.
		w2120869_basketStatus	This attribute records the status of the basket, whether it is ongoing, finalized or archived for future reference.
		w2120869_basketCreatedDate	This attribute records the date that the basket is created.
		w2120869_basketTimeStamp	This attribute records the latest updated time of the basket.

		w2120869_basketRunningTotal	This attribute records the current total of the basket until a member places an order with it.
3	w2120869_Product	w2120869_productId {PK}	This attribute uniquely identifies each product offered by the stores.
		w2120869_productName	This attribute records the name of the product.
		w2120869_productDescription	This attribute records a brief description of each product for the convenience of the members who purchase them.
4	w2120869_Perishable_Product (Specialized)	w2120869_expiaryDate	This attribute records the expiry date of the perishable product.
		w2120869_nutritionalInfo	This attribute records the nutritional info of the perishable product if it is either a food or a drink.
		w2120869_allergenInfo	This attribute records the allergen info of the perishable product for the reference of the members.
5	w2120869_Non_Perishable_Product (Specialized)	w2120869_handlingInstruction	This attribute records the handling instructions for the necessary non perishable products for care when packaging.
		w2120869_warrentyPeriod	This attribute records the duration of warranty for certain non perishable products such as electronics.
6	w2120869_Food (Specialized)	w2120869_dietTags	This attribute records the dietary information of each food product.
7	w2120869_Drink (Specialized)	w2120869_volume	This attribute records the volume contains in each drink product.
		w2120869_brand	This attribute records the brand of each drink product.

8	w2120869_Soft_Drink (Specialized)	w2120869_sugarLevel	This attribute records the sugar level contains in each soft drink.
9	w2120869_Alcohol (Specialized)	w2120869_alcoholPercentage	This attribute records the alcohol percentage contains in each alcoholic drink.
		w2120869_legalMinAge	This attribute records the legal minimum age requires to purchase an alcoholic drink.
10	w2120869_Store	w2120869_storeId {PK}	This attribute uniquely identifies each store which makes it the primary key.
		w2120869_storeName	This attribute records the name of each store.
		w2120869_storeSize	This attribute records the size of each store whether it's a small, medium or a large store.
		w2120869_storeAddress	This attribute records the address of the physical store.
		w2120869_storeTelNo	This attribute records the telephone number of each store for contact purposes.
11	w2120869_Order	w2120869_orderId {PK}	This attribute uniquely identifies each order placed by a member which makes it the primary key.
		w2120869_orderName	This attribute records the name of the order generated by the system once it is confirmed.
		w2120869_orderValue	This attribute records the final value of the order once the order is confirmed.
		w2120869_orderStatus	This attribute records the status of the order whether it is confirmed, ongoing or cancelled.

		w2120869_orderDate	This attribute records the date that the order is placed.
12	w2120869_Payment	w2120869_paymentId {PK}	This attribute uniquely identifies each payment record which makes it the primary key.
		w2120869_paymentName	This attribute records the payment reference given by the members to the payment they made.
		w2120869_paymentType	This attribute records the type of the payment method they chose, whether it is a card payment or mobile payment.
		w2120869_paymentAmount	This attribute records the value of the payment.
		w2120869_paymentTimeStamp	This attribute records the date and the time which the payment was made.
		w2120869_paymentStatus	This attribute records the status of the payment whether it has been successfully paid or not.
13	w2120869_Card_ Payment (Specialized)	w2120869_cardNo	This attribute records the card number of the card used to make the payment.
		w2120869_cardType	This attribute records the type of the card whether it is a credit or a debit card.
		w2120869_cardExpiry	This attribute records the expiry date of the card.
14	w2120869_Mobile_ Payment (Specialized)	w2120869_walletId	This attribute records the id of the mobile wallet used by the member to make the payment.

		w2120869_mobileProvider	This attribute records the mobile provider of the mobile application used to make the payment.
15	w2120869_Collection	w2120869_collectionId {PK}	This attribute uniquely identifies each bundle/collection of orders which makes it the primary key.
		w2120869_collectionName	This attribute records the name given to the collection.
		w2120869_collectionDate	This attribute records the date and time in which the collection is finalized.
		w2120869_collectionStatus	This attribute records the status of the collection whether it is initiated, ongoing or finalized.
		w2120869_expectedDeliveryTime	This attribute records the last expected delivery time of all the orders included in the collection.
		w2120869_collectionCapacity	This attribute records the capacity of the collection whether it is a small collection or a large collection according to the number of orders included in it.
16	w2120869_Delivery	w2120869_deliveryId {PK}	This attribute uniquely identifies each delivery which makes it the primary key.
		w2120869_deliveryName	This attribute records the name given to each delivery.
		w2120869_timeStamp	This attribute records the scheduled date and time of the delivery.
		w2120869_deliveryStatus	This attribute records the status of the delivery whether it is planned, in-route or completed.
		w2120869_totalOrdersIncluded	This attribute records the total orders included in each delivery for

			smooth service making sure that all the placed orders are being delivered to the members in case a collection has missed to record an order included in it which is why this is total orders included in delivery than total collections included.
17	w2120869_Vehicle	w2120869_vehicleId {PK}	This attribute uniquely identifies each vehicle with the ID given to it by the company which makes it the primary key.
		w2120869_registrationNo	This attribute records registration number of each vehicle which is a unique real-world identifier assigned by the transport authority to each vehicle, but it is treated as a regular unique attribute in the conceptual model.
		w2120869_vehicleType	This attribute records the type of vehicle whether it is a two wheel or four wheel.
		w2120869_vehicleCapacity	This attribute records the capacity of each vehicle, how many orders can it carry at an instance.
		w2120869_vehicleStatus	This attribute records the status of vehicle whether it is available or not.
		w2120869_vehicleUsage	This attribute records the used dates and times of each vehicle for any kind of activity.
18	w2120869_Two_ Wheel_Vehicle (Specialized)	w2120869_maxLoadWeight	This attribute records the maximum load of weight that a two wheel vehicle can carry at an instance.
19	w2120869_Four_ Wheel_Vehicle (Specialized)	w2120869_maxVolume	This attribute records the maximum volume that a four wheel vehicle can carry at an instance.

20	w2120869_Worker	w2120869_workerId {PK}	This attribute uniquely identifies each worker in the company which makes it the primary key.
		w2120869_workerName	This attribute records the name of each worker.
		w2120869_workerType	This attribute records the role of each worker whether he is a warehouse worker, driver, admin worker or a cleaner etc.
		w2120869_workerTelNo	This attribute records the telephone number of each worker for contact purposes.
		w2120869_workerEmail	This attribute records the email address of each worker to communicate professional messages.
21	w2120869_Warehouse _Worker (Specialized)	w2120869_shift	This attribute records the shifts of each warehouse worker.
22	w2120869_Driver (Specialized)	w2120869_licensNo	This attribute records the license number of each driver to ensure legal compliance.

Table 6 General and Specialized Entities & their Attributes

B. Relationship Attributes

No.	Relationship Name	Relationship Attributes	Explanation
1	includes	w2120869_quantity	Specifies how many units of a product the member has selected in a particular basket. This varies per basket, not per product.
		w2120869_lineStatus	Indicates the current status of the product line within the basket, ongoing, confirmed, or removed. It

			allows the system to manage basket updates before an order is finalized.
2	offers	w2120869_price	Represents the selling price of a particular product at a specific store. This value may differ between stores depending on location, size, or promotions.
		w2120869_stockLevel	Indicates the number of units available for a specific product at a specific store, which is store specific. Used to check item availability when members add products to their basket.

Table 7 Relationship Attributes

Question 7 – Conceptual EERD

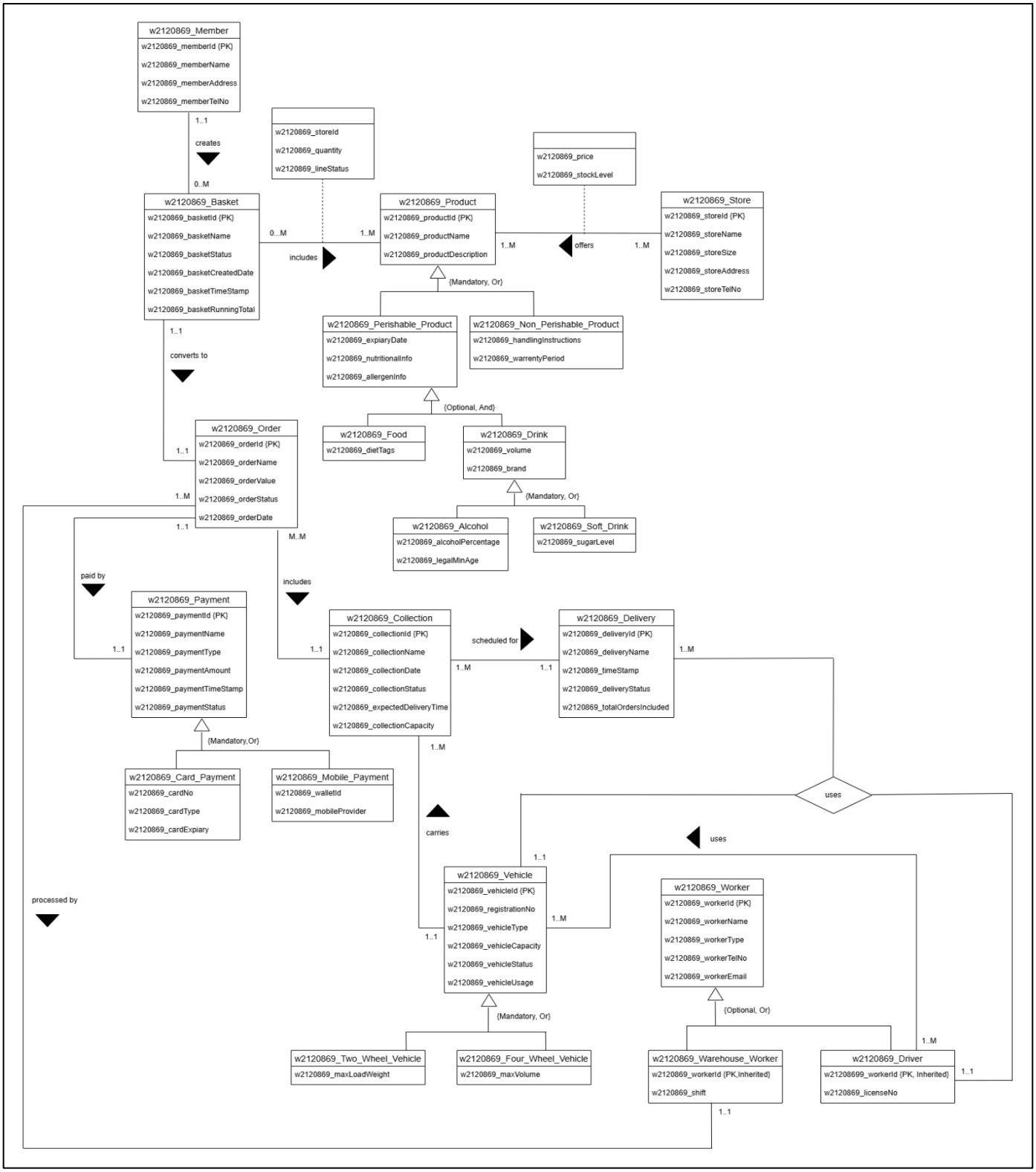

Figure 2 Conceptual EERD

PART B

Question 8 – Mapping Conceptual EERD to Logical ERD

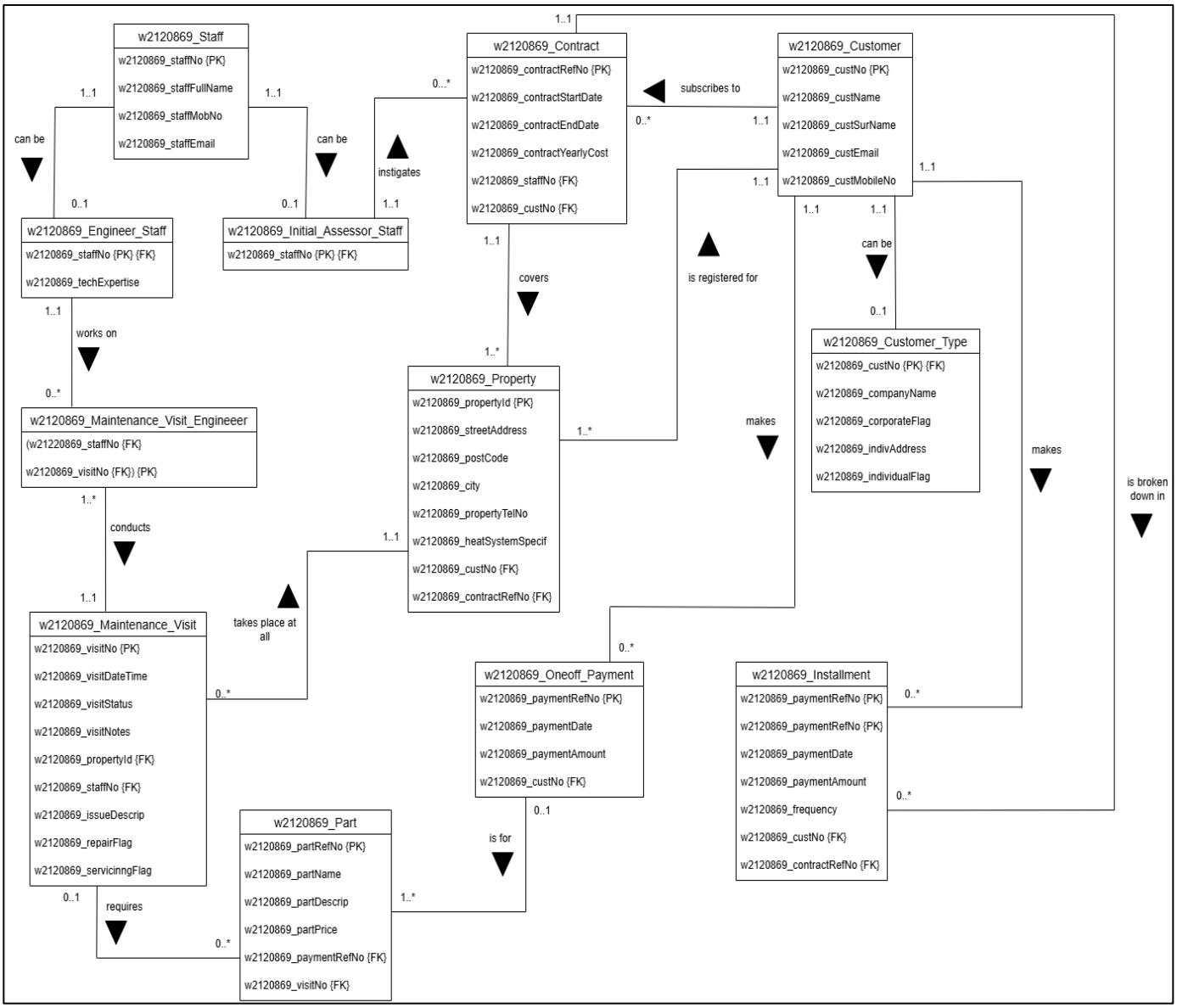

Figure 3 Logical ERD

Question 9 – Mapping Log

No.	Mapping Decision	Justification
1	Mapped the generalized Staff entity table with constraint {Optional, Or} into two separate independent entity tables as Engineer_Staff and Initial_Assessor_Staff.	Created one table for the general entity name **'w2120869_Staff'** as the parent table. Then created two separated tables for the specialized entities named **'w2120869_Engineer_Staff'** and **'w2120869_Initial_Assessor_Staff'** as the child tables according to the mapping rules. Created two 'one-to-one' relationships, optional on one side, between the parent table and the two child tables. Both the PK and FK of each child table is the PK of the parent table which is w2120869_staffNo inherited by the mapping rules.
2	Mapped the generalized Customer entity with constraint {Optional,And} and created a single independent entity table as w2120869_Customer_Type.	Following the mapping strategy created one table for general entity named **'w2120869_Customer'** as the parent table and, **'w2120869_Customer_Type'** for both specialized entities, merged together which becomes the child table. Added flags **'w2120869_corporateFlag'** and **'w2120869_individualFlag'** to differentiate between records of previous specialized entities. Added one 'one-to-one' relationship, optional on the child table side. Both the PK and the FK of the child table is referenced from the PK of the parent table which is w2120869_custNo according to the mapping rules.
3	Mapped the generalized Payment entity with constraint {Mandatory,Or} and created two independent entity tables as w2120869_Oneoff_Payment and w2120869_Installment discarding the Payment entity.	Created 2 separate tables named **'w2120869_Oneoff_Payment'** and **'w2120869_Installment'** according to the mapping rule as, one table for each of the specialized entities with the attributes of the general entity. PK for both tables are the same as the PK of the original general entity which is w2120869_paymentRefNo. The separate relationships associated with the specialized entities stays the same, as the relationships

		that were associated with the general entity doubles up.
4	Mapped the generalized entity Maintenance_Visit with the constraint {Mandatory, And} and combined the two subclasses with the superclass eliminating the subclasses from the logical erd.	Merged all entities into one table with under a new table named **'w2120869_Maintenance_Visit'** according to the mapping rules. Created PK as the same as the generalized entity which is w2120869_visitNo. Added flags as **'w2120869_repairFlag'** and **'w2120869_servicingFlag'** to vary the previous existing records of the specialized entities.
5	Resolved the Many to Many relationship between Maintenance_Visit and Engineer, creating a new entity table as w2120869_Maintenance_Visit_Engineer.	Created a link table named **'w2120869_Maintenance_ Visit_Engineer'** as the child table associated to the two parent tables **'Engineer'** and **'Maintenance_Visit'** through two 'One- to- Many' relationships. FKs of the associated table are the same as the PKs of the parent tables which are w2120869_staffNo and w2120869_visitNo. As for the PK of the associated table used a 'compound key', a combination of the two PKs of the parent tables.
6	Mapped the 'One-to-Many' relationship between the two entities **'Part'** and **'Maintenance_Visit'**	Since the **'Repair'** entity that was in the conceptual model gets merged into **'w2120869_Maintenance_ Visit'** entity, it inherits the relationship with the entity **'Part'**. Hence, added the **'w2120869_Maintenance_Visit'**, the parent table on the 'one' side of the relationship and **'w2120869_Part'**, the child on the 'many' side of the relationship. Created the FK referenced to the PK of the parent table which is w2120869_visitNo.
7	Mapped the 'One-to-Many' relationship between the two entities **'Payment'** and **'Customer'**.	Since the 'Payment' entity that was in the conceptual model gets discarded and the subclasses of that superclass becomes two independent entities in logical erd according to the mapping rules, the One to Many relationship between **'Payment'** and **'Customer'** is inherited by the two entities **'w2120869_Oneoff_Payment'** and **'w2120869_Installment'** with these two being on

		'Many' side of the relationship and w2120869_Customer being on 'One' side of the relationship.
8	Added FKs to the child tables associated with 'One to Many' relationships.	In the conceptual level, FKs are not used. Therefore, adding FKs to the child tables associated with One to Many relationships are essential. Created FK of the child table which is on 'Many' side of the relationship as a copy of the PK of the parent table which is on 'One' side. FK of the child table references the PK of the parent table. In the Logical erd 'w2120869_Customer' has a relationship with 'w2120869_Contract' as 'subscribed to' with the multiplicity of contract side being '0..*' and customer side being '1..1'. Hence this is a 'One to Many' relationship and 'w2120869_Contract' being on 'Many' side, it has inherited the PK of the parent table Customer which is 'w2120869_custNo' as FK as per the mapping rules.

Table 8 Mapping Log